Ericsson PE Node WFD-PE-02 Standard Operating Procedure for Major Hardware/Environmental Faults

Node: WFD-PE-02 | **Vendor:** Ericsson | **Severity:** Major | **Alarm Code:** ALARM_1542

Description

This SOP outlines the procedures to be followed in case of major hardware/environmental faults on the Ericsson PE node WFD-PE-02 at the Whitefield POP. The failure domain of interest is hardware/environmental faults, specifically ASIC memory parity errors, line card crashes, and power supply redundancy loss. The PE node is a critical component of the network, responsible for providing IP/MPLS services to customers. The Ericsson CLI dialect is used for diagnostic and resolution commands due to the vendor-specific nature of the equipment. The underlying protocol and hardware mechanics that contribute to these failures include the use of ASICs with internal memory parity checks, which can fail due to physical faults or environmental stressors. Line card crashes can occur due to overheating, power supply issues, or software faults. The power supply redundancy loss can lead to a single point of failure, resulting in network downtime. Understanding these root causes is essential to identifying and resolving the issues efficiently.

Pre-requisite Checks

- Verify that the remote hands have been dispatched to the Whitefield POP for assistance.
- Check the status of the redundant links to ensure they are operational.
- Ensure that the Ericsson CLI is accessible and functional on the PE node.

Potential Root Causes

Cause	Probability
ASIC memory parity error	High
Line card crash due to overheating	Medium
Power supply redundancy loss	Low

Diagnostic Commands & Expected Outputs

```
Command: show system redundancy, Expected Output: Redundancy information for the system, including the status of the power supplies and line cards., Description: This command provides an overview of the redundancy status of the system, helping to identify any potential issues with the power supplies or line cards.

Command: show lcu status, Expected Output: Status information for the line cards, including any alarms or errors., Description: This command provides detailed status information for the line cards, including any alarms or errors that may be contributing to the failure.
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```
Command: show asic status, Expected Output: Status information for the ASICs, including any errors or alarms., Description: This command provides detailed status information for the ASICs, including any errors or alarms that may be contributing to the failure.
```

Resolution Steps

Step 1: Step: Replace the failed line card, Command: replace lcu , Description: Replace the failed line card with a new one, ensuring that the replacement is verified and confirmed to be operational.

Step 2: Step: Replace the failed ASIC, Command: replace asic , Description: Replace the failed ASIC with a new one, ensuring that the replacement is verified and confirmed to be operational.

Step 3: Step: Verify power supply redundancy, Command: verify power supply redundancy, Description: Verify that the power supply redundancy is operational, ensuring that there is no single point of failure.

Post-Check Verification

```
Command: show system redundancy, Expected Output: Redundancy information for the system, including the status of the power supplies and line cards., Description: Verify that the system redundancy is operational, ensuring that there are no issues with the power supplies or line cards.

Command: show lcu status, Expected Output: Status information for the line cards, including any alarms or errors., Description: Verify that the line cards are operational, ensuring that there are no alarms or errors.

Command: show asic status, Expected Output: Status information for the ASICs, including any errors or alarms., Description: Verify that the ASICs are operational, ensuring that there are no errors or alarms.
```

Roll-back Procedures

- Step: Restore the original configuration, Command: restore original configuration, Description: Restore the original configuration to ensure that the system returns to its previous state.

- Step: Verify system functionality, Command: verify system functionality, Description: Verify that the system is functioning as expected, ensuring that the changes made during the troubleshooting process did not introduce any new issues.